Numerical Integration: Trapezoidal and Simpson Rules

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Abstract

This note presents two simple methods for computing definite integrals numerically: the trapezoidal rule and Simpson's rule. We state their error bounds, test both rules on a single example, and compare the observed errors with the theoretical rates.

1 Introduction

Many integrals cannot be expressed with elementary functions. For example, the integral of e^{-x^2} over $[0, 1]$ has no closed form in elementary terms, so numerical methods are needed. Given a function f on an interval $[a, b]$, we want to approximate

$$I = \int_a^b f(x)\,\mathrm{d}x. \tag{1}$$

2 The trapezoidal rule

Divide $[a, b]$ into n equal subintervals of width $h = (b-a)/n$, with nodes $x_i = a+ih$. The *trapezoidal rule* is

$$T_n = h\left[\frac{f(a)}{2} + \sum_{i=1}^{n-1} f(x_i) + \frac{f(b)}{2}\right]. \tag{2}$$

Proposition 1 (Error of the trapezoidal rule)**.** *If $f \in C^2[a, b]$, then*

$$|I - T_n| \le \frac{(b-a)\,h^2}{12} \max_{x\in[a,b]} |f''(x)|. \tag{3}$$

3 Simpson's rule

If n is even, *Simpson's rule* is

$$S_n = \frac{h}{3}\Big[f(x_0) + 4\sum_{\substack{i=1\\ i \text{ odd}}}^{n-1} f(x_i) + 2\sum_{\substack{i=2\\ i \text{ even}}}^{n-2} f(x_i) + f(x_n)\Big]. \tag{4}$$

Proposition 2 (Error of Simpson's rule)**.** *If $f \in C^4[a, b]$, then*

$$|I - S_n| \le \frac{(b-a)\,h^4}{180} \max_{x\in[a,b]} |f^{(4)}(x)|. \tag{5}$$

4 Numerical experiment

We take $f(x) = e^{-x^2}$ on $[0, 1]$. The exact value is $I = \frac{\sqrt{\pi}}{2}\,\mathrm{erf}(1) \approx 0.7468241328$. Figure 1 shows the function and the four trapezoids used for $n = 4$, and Table 1 lists the approximations and their errors.

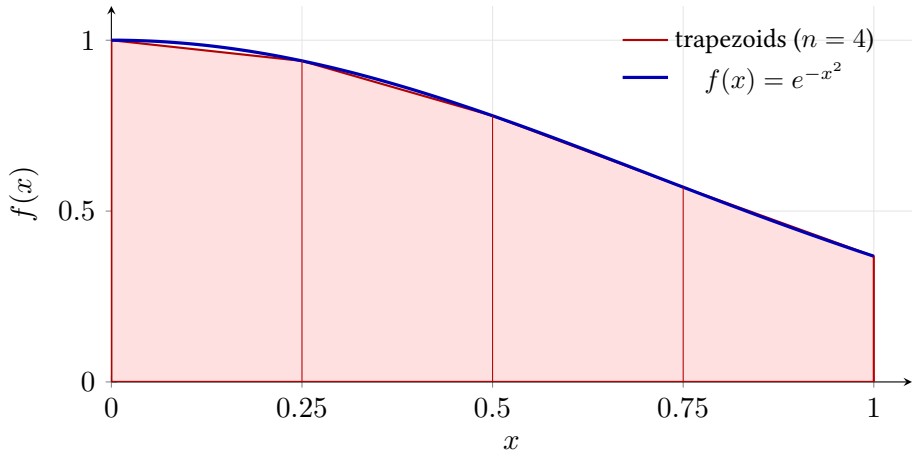


Figure 1: The function $f(x) = e^{-x^2}$ on $[0, 1]$ and the four trapezoids of the rule T_4.

Table 1: Approximations and absolute errors for $\int_0^1 e^{-x^2}\,\mathrm{d}x$.

n	T_n	S_n	$\lvert I - T_n\rvert$	$\lvert I - S_n\rvert$
2	0.731 370	0.747 180	1.545e−2	3.563e−4
4	0.742 984	0.746 855	3.840e−3	3.125e−5
8	0.745 866	0.746 826	9.585e−4	1.988e−6
16	0.746 585	0.746 824	2.395e−4	1.246e−7

5 Conclusion

Simpson's rule is much more accurate than the trapezoidal rule for this smooth integrand. When n is doubled, the error of the trapezoidal rule is divided by about 4 (order h^2), whereas the error of Simpson's rule is divided by about 16 (order h^4), in agreement with Eqs. (3) and (5). For further reading, see [1, 2, 3].

References

[1] Richard L. Burden, J. Douglas Faires, and Annette M. Burden. *Numerical Analysis*. 10th ed. Boston: Cengage Learning, 2016.

[2] Kendall E. Atkinson. *An Introduction to Numerical Analysis*. 2nd ed. New York: John Wiley & Sons, 1989.

[3] William H. Press et al. *Numerical Recipes: The Art of Scientific Computing*. 3rd ed. Cambridge: Cambridge University Press, 2007.